

```

1  (*=====
2  Readings:
3
4      This lab has no prerequisite textbook reading.
5
6  =====*)
7
8  (*
9
10         CS51 Lab 20
11         Synthesis
12
13 *)
14 type image = float list list ;;
15 (* images are lists of lists of floats between 0. (white) and 1. (black) *)
16 type size = int * int ;;
17 open Graphics ;;
18
19 (* threshold threshold image -- image where pixels above the threshold
20 value are black *)
21 let threshold img threshold =
22     List.map (fun row -> List.map (fun v -> if v <= threshold then 0. else 1.)
23              row) img
24
25 (* show the image *)
26 let depict img =
27     Graphics.open_graph ""; Graphics.clear_graph ();
28     let x, y = List.length (List.hd img), List.length img in Graphics.resize_window x y;
29     let depict_pix v r c = let lvl = int_of_float (255. *. (1. -. v)) in Graphics.set_color
30     plot c (y - r) in
31     List.iteri (fun r row -> List.iteri (fun c pix -> depict_pix pix r c) row) img;
32     Unix.sleep 2; Graphics.close_graph ();
33
34 (* dither max image -- dithered image *)
35 let dither img =
36     List.map
37       (fun row ->
38         List.map
39           (fun v -> if v > Random.float 1.
40                    then 1.
41                    else 0.) row)
42       img
43
44 let mona = Monalisa.image ;;
45
46 depict mona ;;
47
48 let mona_threshold = threshold mona 0.75 ;;
49 depict mona_threshold ;;
50
51 let mona_dither = dither mona ;;
52 depict mona_dither ;;

```